

Passive cardiac shear-wave elastography, part 2: smoothing, M-line averaging, SVD and CFWI on the velocity 15–150 Hz default

shearWaveProcessing – study/analysis/passive_methods_atlas.py --set v2

24 September 2026

What changed since part 1

Part 1 ([passive_methods.pdf](#)) compared the processing choices on a displacement base (view A). It pointed to **velocity with a band-pass starting at ~15 Hz** as the most readable space-time. This part makes that the default and re-evaluates, one at a time:

- spatial smoothing;
- temporal smoothing;
- M-line averaging, and the three together;
- the SVD clutter filter;
- clutter filter wave imaging (CFWI).

New default for this part. Loupas lag-1 frame-to-frame **velocity**; zero-phase Butterworth band-pass **15–150 Hz**; Gaussian $\sigma = 0.6$ mm axial $\times$ 1.2 mm lateral; moving mean 3; 5 M-lines 0.5 mm apart; no directional filter.

The same seven in-vivo windows and panel conventions as part 1 are used:

- x = time relative to the detected event peak (ms), y = distance along the analysed part of the M-line (mm);
- colour scaled per panel;
- six windows scored *clear* (three AVC, three MVC), and the last column scored *none*, which shows what a setting makes of a window without a visible wave;
- no automatic speeds.

Auxiliary numbers (not a speed)

Each recipe also gets two numbers (study/logs/passive_atlas_v2.csv):

- **Hand-line tracking** (median and minimum over the six clear windows): the mean $|\text{signal}|$ sampled along your hand-drawn velocity wavefront, divided by the panel RMS. About 1 means the line is no better than noise; higher means the wavefront stands out.
- **No-wave best line**: the same score for the *strongest* straight line the no-wave panel offers (normalised Radon). It measures how much line-like structure a setting invents where there is no wave.

Two caveats:

- Smoothing raises *both* numbers, so only the gap between them is informative.
- The hand lines were drawn on velocity panels (view C, 15–90 Hz), which favours recipes close to that. They are a guide, not a verdict; the plots are the evidence.

family	recipe	hand line, median	hand line, min	no-wave best line
spatial	none	2.00	1.59	1.02
	Gaussian 0.3×0.6 mm	2.14	1.84	1.22
	Gaussian 0.6×1.2 mm (default)	2.24	2.03	1.52
	Gaussian 1.0×2.0 mm	2.30	2.09	1.92
	Gaussian 1.5×3.0 mm	2.41	2.14	2.00
	Gaussian 2.0×4.0 mm	2.48	2.18	1.86
	median 1.0×2.0 mm	2.21	1.88	1.37
temporal	none	2.22	1.99	1.52
	moving mean 3 (default)	2.24	2.03	1.52
	moving mean 5	2.26	2.08	1.48
	moving mean 7	2.28	2.15	1.38
	moving mean 9	2.22	2.01	1.29
	moving median 5	2.22	2.02	1.48
	Savitzky–Golay 9, order 3	2.25	2.02	1.51
M-lines	1 line	2.19	1.97	1.39
	3 lines $\times 0.5$ mm	2.21	2.00	1.45
	5 lines $\times 0.5$ mm (default)	2.24	2.03	1.52
	9 lines $\times 0.5$ mm	2.28	2.06	1.57
	9 lines $\times 0.8$ mm	2.33	2.10	1.52
	15 lines $\times 0.5$ mm	2.33	2.10	1.52
	15 lines $\times 0.8$ mm	2.32	2.16	1.47
combined	none (no spatial, no temporal, 1 line)	1.73	1.13	0.88
	light (Gauss 0.3×0.6 , mean 3, 3 lines)	2.07	1.74	1.21
	default	2.24	2.03	1.52
	medium (Gauss 1.0×2.0 , mean 5, 9 lines)	2.37	2.15	1.80
	heavy (Gauss 2.0×4.0 , mean 9, 15 lines)	2.39	2.22	1.95
SVD	none (default)	2.24	2.03	1.52
	IQ SVD, 1 component	2.23	2.16	1.40
	IQ SVD, 2 components	2.22	1.89	1.84
	IQ SVD, 3 components	2.23	1.94	1.96
	velocity-field SVD, 1 component	2.20	0.38	1.49
	IQ SVD 2 instead of the high-pass	1.60	1.26	1.54

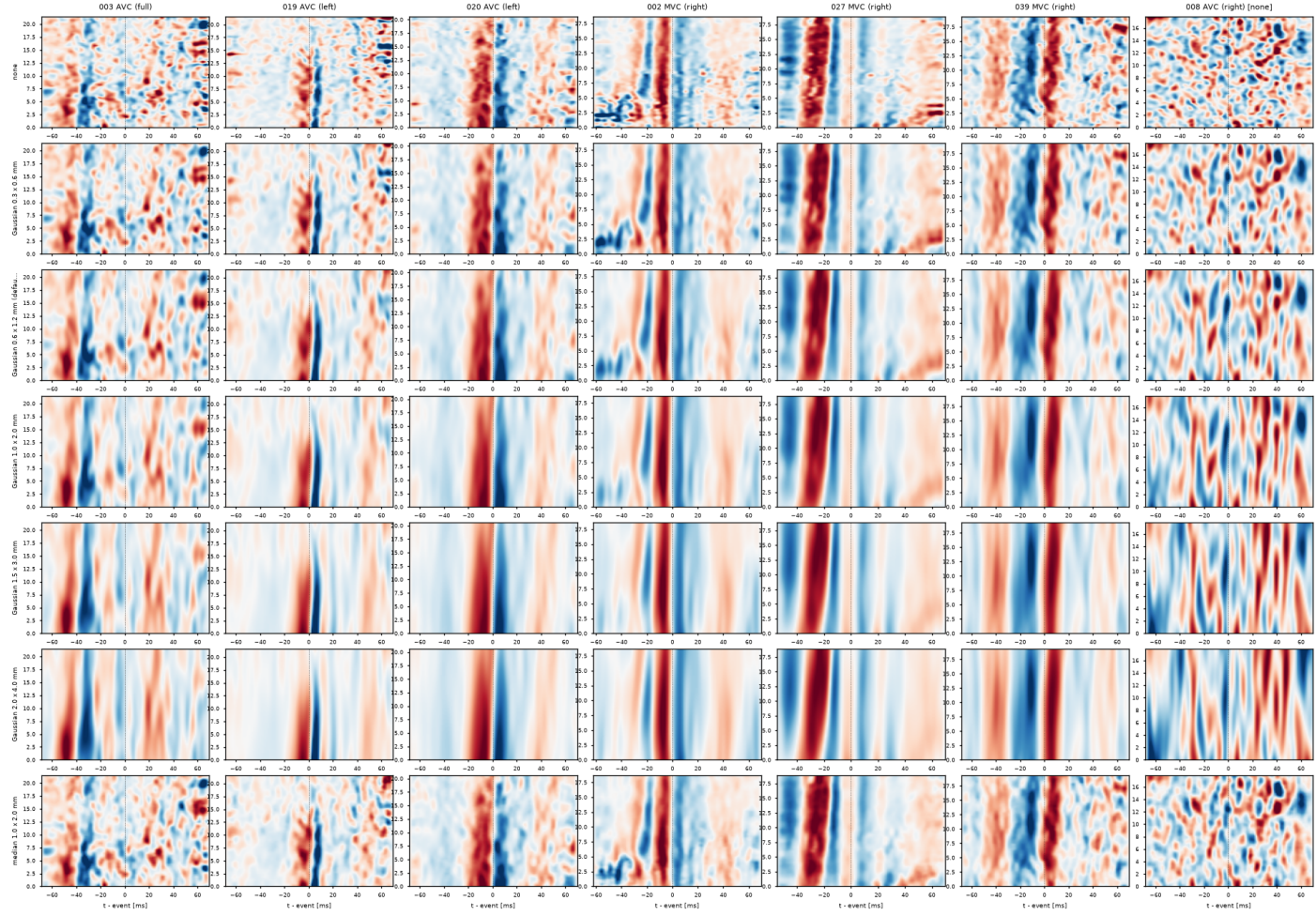
CFWI is left out of the table: its map does not follow the velocity wavefront your lines were drawn on (hand-line scores 0.45–0.75), so the score does not apply. It is judged on the plots.

1 Spatial smoothing

Per-frame smoothing of the velocity field before the M-line is sampled.

- **None:** the fronts are there but carry speckle texture, and the no-wave window is pure speckle.
- **Gaussian 0.3×0.6 mm** removes part of it.
- **Gaussian 0.6×1.2 mm (default)** gives clean fronts in all six windows.
- **Gaussian 1.0×2.0 mm** is slightly cleaner again (003, 019, 002).
- **From 1.5×3 mm**, the real fronts hardly improve while the no-wave window turns into long, clean-looking tilted and vertical bands. The no-wave best line rises from 1.5 to 2.0 and closes most of the gap to the real fronts.
- **Median 1×2 mm** keeps more texture than the Gaussian of the same size.

Reading: Gaussian between 0.6×1.2 and 1.0×2.0 mm; beyond that, smoothing starts to create the structure it is supposed to reveal.

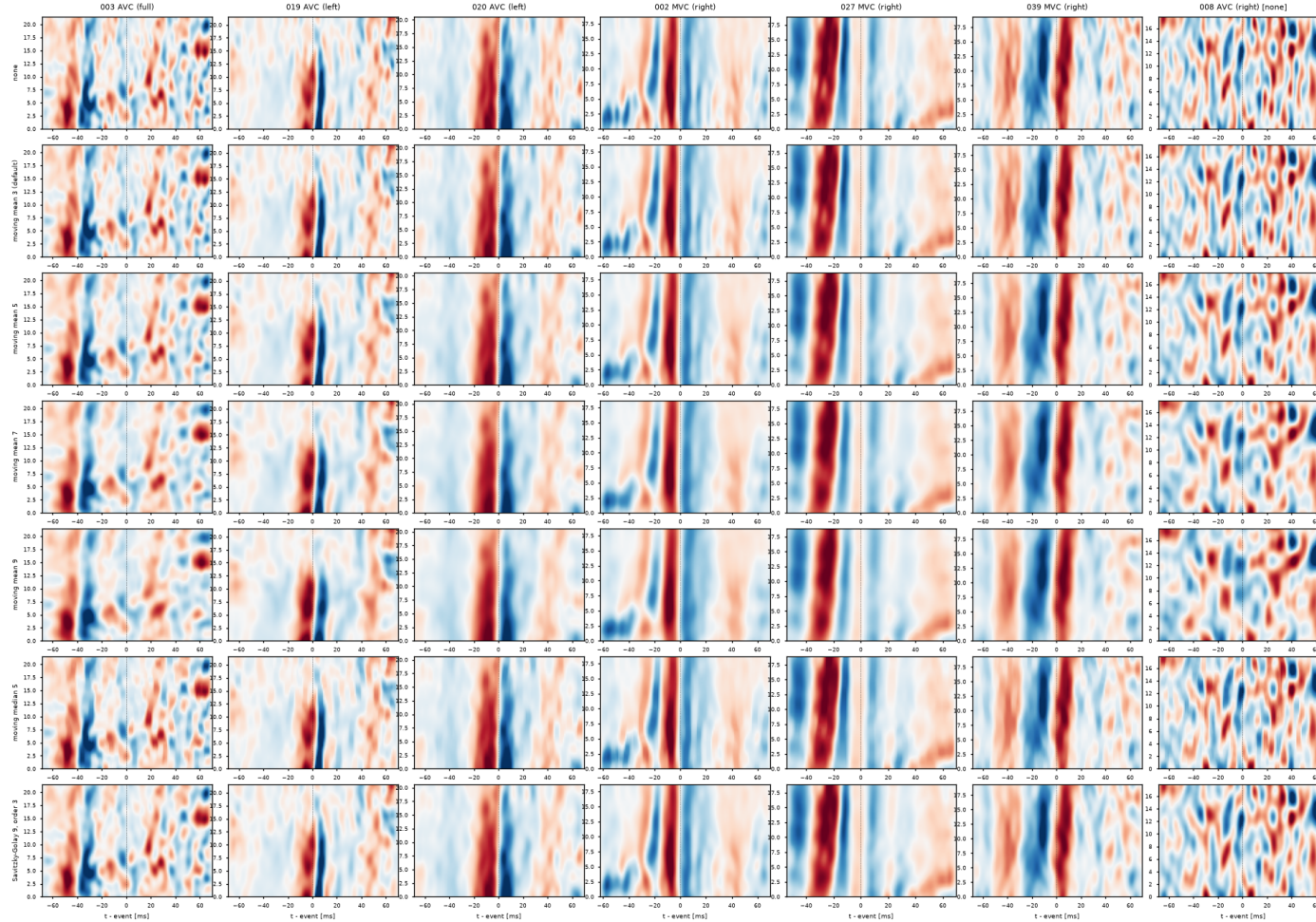


2 Temporal smoothing

A moving mean, median or Savitzky–Golay filter over 3–9 frames (3–10 ms).

- Up to a mean of 5 the panels barely change. Median 5 and Savitzky–Golay 9 look like the default.
- Means of 7 and 9 visibly widen the fronts (019, 002, 039). They also make the no-wave window coarser but not more line-like (best line $1.52 \rightarrow 1.29$): unlike spatial smoothing, temporal smoothing does not invent straight lines here.

Reading: mean 3–5; longer windows cost front sharpness for little gain.

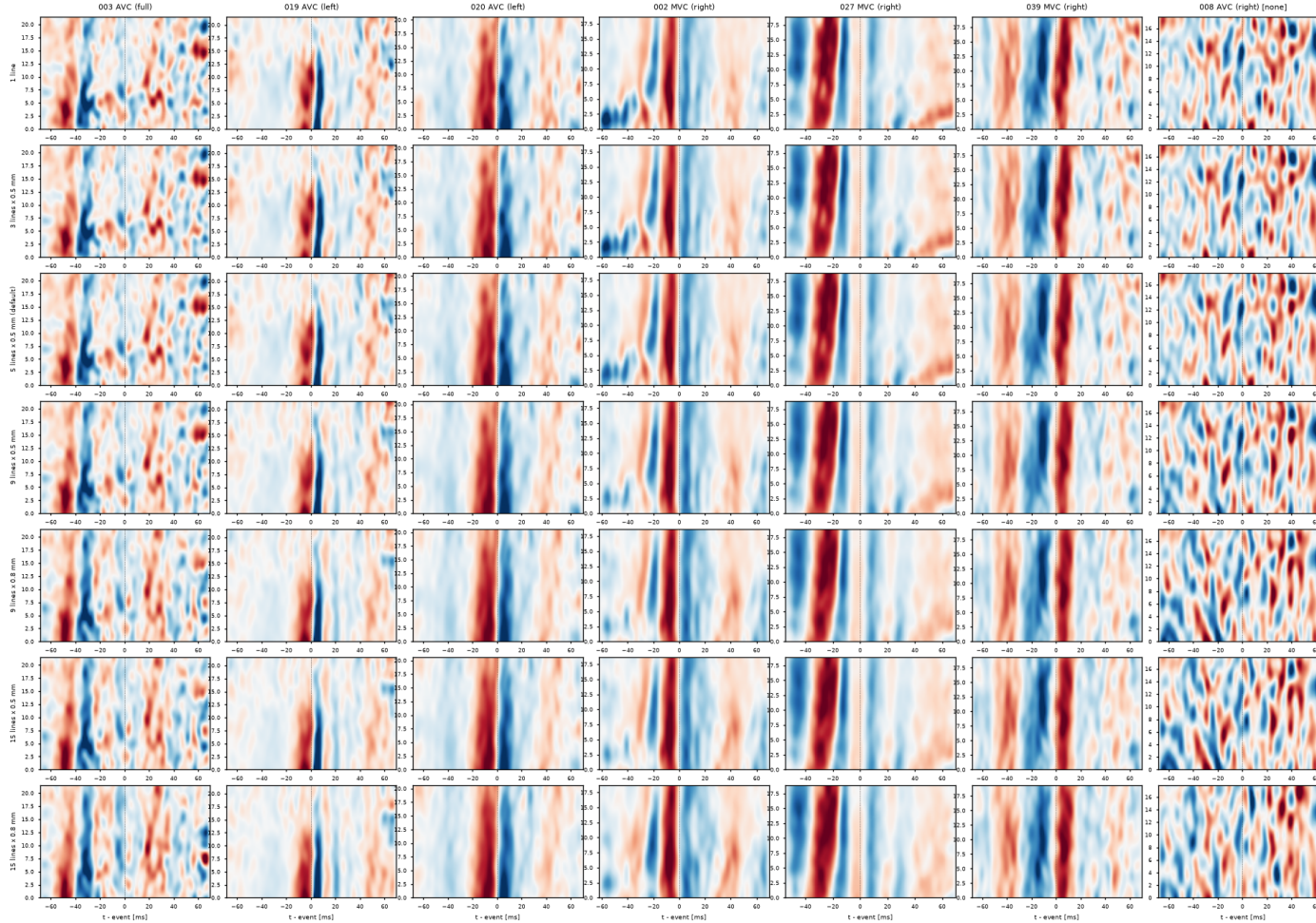


3 M-line averaging

Parallel copies of the M-line along its normal, averaged. The wall is about 8–12 mm thick: 9 lines $\times$ 0.5 mm span 4 mm, and 15 lines $\times$ 0.8 mm span 11 mm.

- 1 to 5 lines look alike.
- 9 lines $\times$ 0.5 mm clean the fronts a little further (003, 002).
- Wider spans (9 $\times$ 0.8, 15 lines) smooth further. In the no-wave window they start to form diagonal stripes, as the averaging band reaches the wall edges and blood.

Reading: 5–9 lines at 0.5 mm, a band of 2–4 mm kept inside the wall.

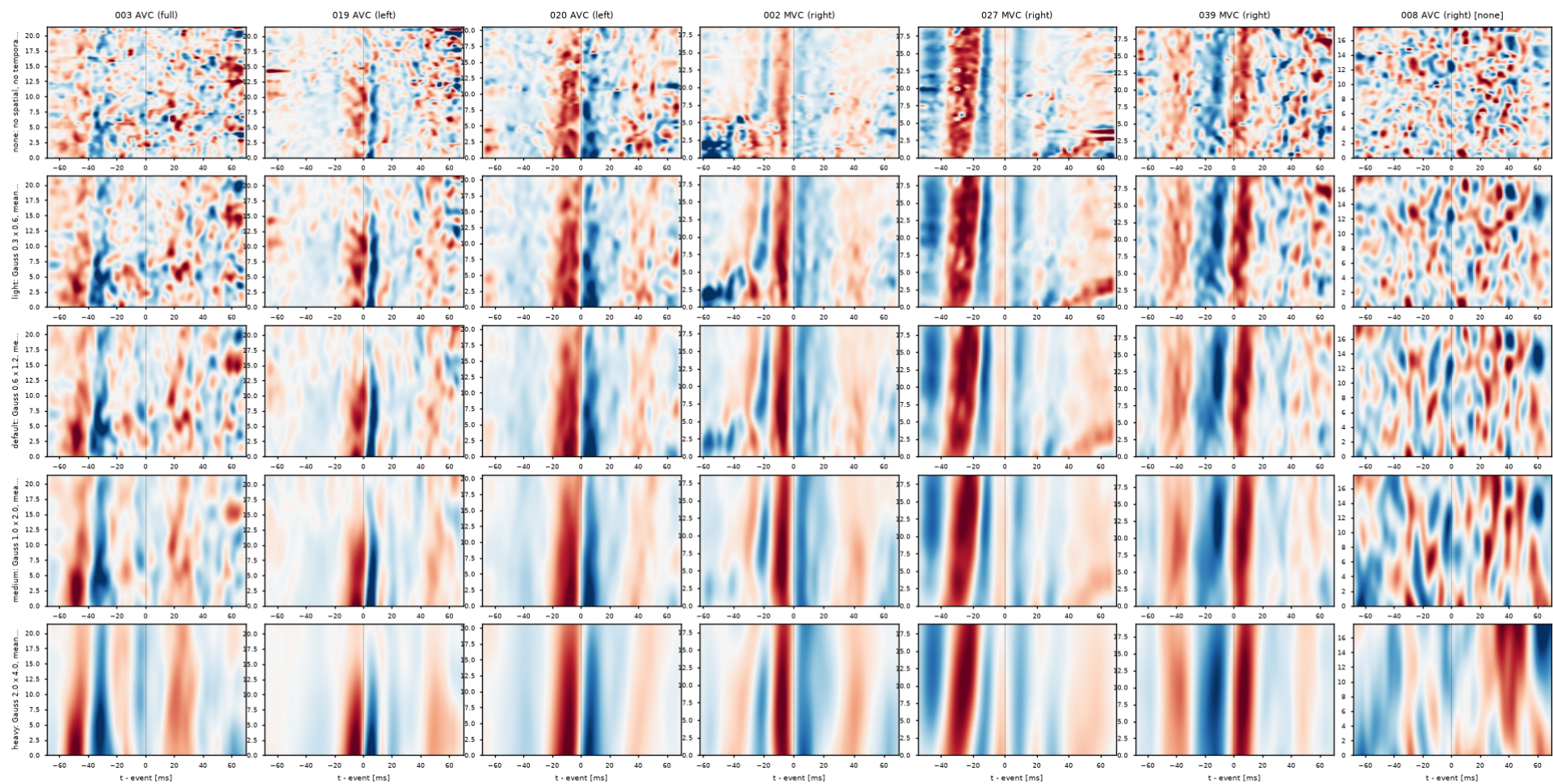


4 The three together

The same three settings changed together, from none to heavy.

- **None** is speckle-dominated (hand-line minimum 1.13).
- **Light** is still noisy.
- **Default** is clean.
- **Medium** (Gauss 1.0×2.0 , mean 5, 9 lines) gives the cleanest fronts while the no-wave window stays irregular.
- **Heavy** makes everything smooth, widens the fronts and turns the no-wave window into broad bands.

Reading: default to medium. Medium is the best-looking setting on these seven windows. Its no-wave score is higher (1.80 vs 1.52), so it should be checked on more windows before it becomes the default.

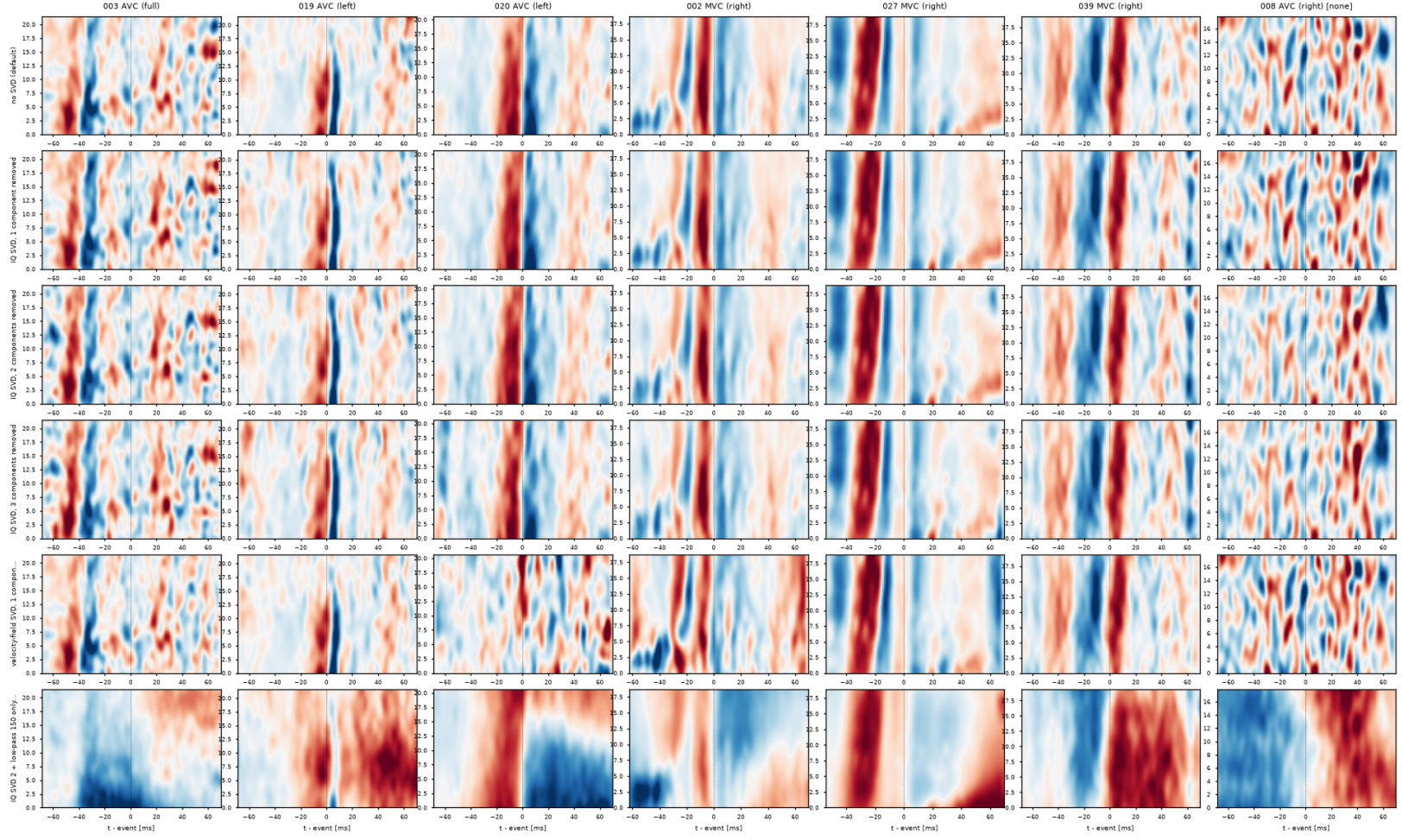


5 SVD clutter filtering

Singular-value decomposition removing the dominant slow-time components: on the IQ (Demené et al. 2015) before estimation, or on the velocity field. Part 1 found it inert on a displacement base. Re-tested on velocity:

- **1 IQ component:** no visible change. The worst window improves slightly (hand-line minimum $2.03 \rightarrow 2.16$) and the no-wave score drops ($1.52 \rightarrow 1.40$). Harmless, possibly marginally useful.
- **2–3 IQ components:** fronts intact, but more background texture (003, 020) and a more line-like no-wave window (1.84–1.96). This starts removing tissue signal.
- **SVD on the velocity field: destroys the wave in 020** (replaced by noise; hand-line minimum 0.38) and distorts 002. Reject.
- **IQ SVD instead of the high-pass:** broad slow drifts dominate every panel. SVD does not replace the 15 Hz high-pass, as part 1 found for displacement.

Reading: not needed. If used, IQ only, 1 component, and never on the field or instead of the band-pass.

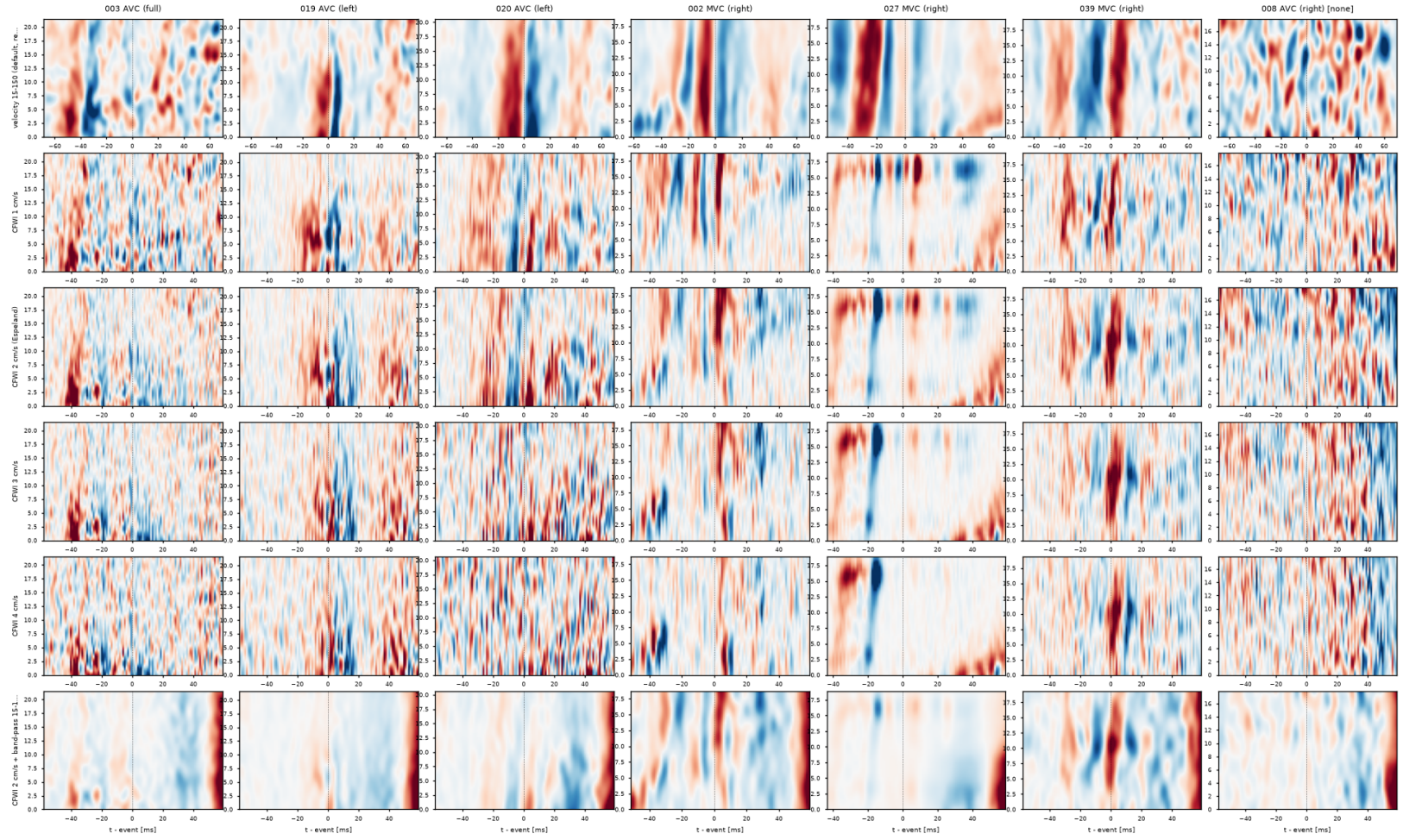


6 Clutter filter wave imaging (CFWI)

Espeland et al. (2024): slow-time IQ high-pass at a cutoff velocity, envelope, temporal derivative, then the default smoothing. 10 frames are dropped at each window end (filter transient). At our 3.9 MHz demodulation and 926 Hz frame rate, 1–4 cm/s is a 51–203 Hz cutoff against a 9.1 cm/s aliasing limit.

- At every cutoff the map is a streaky, vertically striped texture.
- The event shows as a front in 002, 027 and 039 (clearest at 3–4 cm/s), faintly in 019, and hardly in 003 and 020.
- The no-wave window has the same streak texture as the real ones, which makes a real front harder to tell apart.
- Band-passing the CFWI map (last row) adds a large end artifact.
- In the benchmark on 15 windows its automatic direction agreed with the hand picks only at chance level.

Reading: on these acquisitions CFWI does not approach the velocity map (first row). Why is not established. One untested hypothesis is the frame rate: at 926 Hz the 51–203 Hz cutoffs above take up a large part of the 463 Hz slow-time bandwidth, so the filter’s transition band covers much of the tissue-velocity range rather than sitting at its edge.



7 Summary

choice	what the plots and numbers show	suggested default
spatial smoothing	none: speckle; 0.6–1.0 mm Gaussian: clean fronts; ≥ 1.5 mm: the no-wave window turns into bands	Gaussian 0.6 $\times$ 1.2 to 1.0 $\times$ 2.0 mm
temporal smoothing	little change up to mean 5; mean 7–9 widen fronts; no invented lines	moving mean 3–5
M-line averaging	1–5 lines alike; 9×0.5 mm slightly cleaner; wide bands reach the wall edges	5–9 lines $\times$ 0.5 mm
combined	medium (1.0×2.0 , mean 5, 9 lines) looks best but raises the no-wave score	default now; medium after checking more windows
SVD	1 IQ component harmless; more components or field SVD damage the wave; cannot replace the high-pass	off (or IQ, 1 component)
CFWI	streaky; front visible in 3–4 of 6 windows; worse than velocity	not used

A concrete candidate for configs/passive.yaml view A:

- velocity, band-pass 15–150 Hz;
- Gaussian 0.6–1.0 mm $\times$ 1.2–2.0 mm;
- moving mean 3–5;
- 5–9 M-lines $\times$ 0.5 mm;
- no directional filter, no SVD.

This is close to the current view C (velocity, 15–90 Hz, Gauss 1.0×2.0 , mean 5, 9 lines $\times$ 0.8 mm) with a wider band and a narrower M-line band.

Caveats.

- Seven windows, labelled by one observer; the hand lines come from view C panels.
- Colour is scaled per panel.
- The fronts are nearly vertical on 19–22 mm lines, so “cleaner” partly means “more uniform along the line” – the no-wave column is there to keep that in check.
- Before changing the default, run the candidate on the full labelled set (15 windows) and have a second observer read it (docs/TODO.md, items 4 and 11).

Reproduce.

- `python study/analysis/passive_methods_atlas.py --set v2 (~20 min, reads Z:/raw_data); --figures` re-renders from the cache.
- `tectonic passive_methods_v2.tex`.